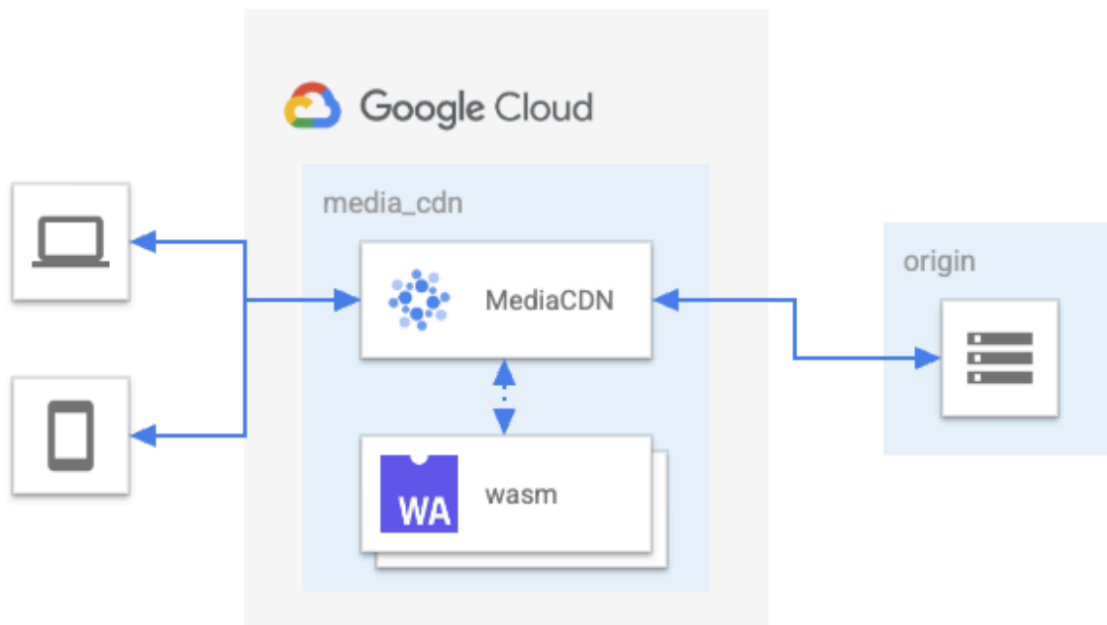


Using Service Extension in Media CDN

Service Extensions for Media CDN, sometimes also referred as Edge Programmability, allows you to run your own code at the edge to customize the behavior of Media CDN. This unlocks additional use cases ranging from normalizing cache key, custom token authentication and token revocation, additional custom log fields, A/B testing, and custom error page. In this case, we will use the service extensions feature to authenticate the signed URL for customers to replace Lambda@Edge in AWS.

MediaCDN Request Flow w/ Service Extensions



1. Setup Environment

1.1 [Install the Rust toolchain.](#)

Using Cloud Shell/VMs, to install the rust tools. At the end of the installation process, follow any instructions printed to the console to finish the configuration process.

1.2 Add Wasm support to the Rust toolchain.

None

```
rustup target add wasm32-wasip1
```

1.3 Install c compiler

None

```
sudo apt update
sudo apt install build-essential
sudo apt install docker.io
```

2. Create custom code by using one of the Proxy-Wasm SDKs

2.1 Preparing the plugin code

You can follow the guide to prepare the plugin code as below.

<https://cloud.google.com/service-extensions/docs/prepare-plugin-code>.

Or download the zip code, and unzip into Cloud Shell.

[my-wasm-plugin.zip](#)

3. Update/Compile your code into a Wasm module.

Update the following parameters in src/[lib.rs](#), use the signed key/backup key and expired seconds.

None

```
let mi_key = String::from("sign-key-mi");  
let mi_key_backup = String::from("sign-key-mi-bak");  
let mi_timestamp_before:u64 = 3600;  
let mi_timestamp_after:u64 = 604800;
```

Compile the code into wasm.

None

```
cargo build --release --target wasm32-wasi
```

4. Build a Docker container image that contains your Wasm module.

None

```
docker build --no-cache --platform wasm -t my-wasm-plugin .
```

5. Publish your container image to an Artifact Registry repository (create a new one if needed.). Specify image tags or digests, as required.

None

```
docker tag my-wasm-plugin  
us-central1-docker.pkg.dev/frankie1-422709/wasm/my-wasm-plugin:prod  
  
docker push us-central1-docker.pkg.dev/frankie1-422709/wasm/my-wasm-plugin:prod
```

6. Create a plugin that contains the uploaded wasm module, replace the image url pushed above.

None

```
gcloud alpha service-extensions wasm-plugins create my-wasm-plugin-resource
--log-config=enable=true,sample-rate=1.0,min-log-level=DEBUG

gcloud alpha service-extensions wasm-plugin-versions create my-version-1
--wasm-plugin=my-wasm-plugin-resource
--image="us-central1-docker.pkg.dev/frankie1-422709/wasm/my-wasm-plugin:prod"

gcloud alpha service-extensions wasm-plugins update my-wasm-plugin-resource
--main-version=my-version-1

gcloud alpha service-extensions wasm-actions create my-wasm-action-resource
--wasm-plugin=my-wasm-plugin-resource
```

7. Configure the Media CDN following the guide below.

CDN Configuration Guide:

<https://cloud.google.com/media-cdn/docs/configuration>

SSL configuration guide:

<https://cloud.google.com/media-cdn/docs/ssl-certificates>

Notes: SSL certificate for cache validation needs more than 10 minutes.

8. Configure your Media CDN service to use the required host rules and route rules, and link the plugin to it.

Download the service.yaml from media cdn.

None

```
gcloud edge-cache services export media-cdn --destination=my-service.yaml
```

Open the my-service.yaml file and add a wasmAction to the routeAction for the given route, which references the WasmPlugin resource created earlier.

None

```
logConfig: {}
name:
projects/nimble-unison-325208/locations/global/edgeCacheServices/media-cdn
routing:
  hostRules:
  - hosts:
    - '*'
    pathMatcher: path-matcher-0
  pathMatchers:
  - name: path-matcher-0
    routeRules:
    - headerAction: {}
      matchRules:
      - prefixMatch: /
      origin:
projects/256146349297/locations/global/edgeCacheOrigins/cloud-storage-origin
  priority: 1
  routeAction:
    cdnPolicy:
      cacheKeyPolicy: {}
      cacheMode: FORCE_CACHE_ALL
      defaultTtl: 3600s
      signedRequestMode: DISABLED
    wasmAction: "my-wasm-action-resource"
```

Import the media cdn.

None

```
gcloud alpha edge-cache services import media-cdn --source=my-service.yaml
```

<https://mediacd.hongbocheng.demo.altostrat.com/V12.5.8.0.RKPCNXM/miui-blockota-chopin-V12.5.7.7.RKPCNXM-V12.5.8.0.RKPCNXM-bbd982b688-11.0.zip?t=1721035765&s=9c32c66c87b053844f3fe3d44f587e38>

/V12.5.8.0.RKPCNXM/miui-blockota-chopin-V12.5.7.7.RKPCNXM-V12.5.8.0.RKPCNXM-bbd982b688-11.0.zipsign-key-mi1728953470

Replace the timestamp to current date/time.